6

Computer Vision (CV)

Computer vision is a field of artificial intelligence that helps computers gain a high level of understanding from digital images or videos. It is done by acquiring images, processing them, analysing them and finally understanding the information they are conveying. If AI allows computers to think, computer vision helps them to see, observe and understand.

6.1 Human Vision vs Computer Vision

Computer vision works like human vision. The main advantage of human vision is that we have a brain that automatically learns to tell how far two objects are, whether they are moving or not, whether there is something wrong in an image and many more similar questions.

Computer vision techniques, in turn, train machines to perform these functions. The main challenge here is that machines have to analyse images coming from cameras using data and algorithms in a very limited time (refer **Fig. 6.1**). Once a system is trained to process thousands of images in a minute, it can quickly surpass human capabilities.

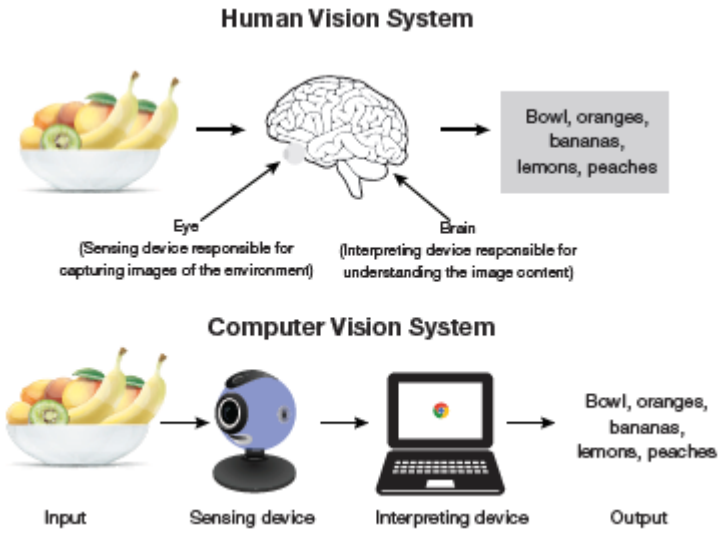


FIGURE 6.1 Human Vision vs Computer Vision

6.2 How Does Computer Vision Work?

Computer vision algorithms need large volume of data for analysing and recognizing images. For this, computer vision makes extensive use of deep learning (next level of machine learning) and Convolutional Neural Network (CNN).

Machine learning or deep learning algorithms help machines learn about the context of visual data and learn by itself how to identify and differentiate images. A CNN complements the machine learning or deep learning model by breaking images down into pixels that are given tags or labels. The CNN uses these labels to perform convolutions (a mathematical operation on two functions to produce a third function) to make predictions about what it is 'seeing.' Multiple iterations of the neural network are run and the accuracy of their predictions is evaluated. The process is repeated until the computer identifies the image with an acceptable accuracy. At this time, we say that the machine is recognizing or seeing images in a way similar to humans.

6.3 The Evolution of Computer Vision

Prior to deep learning/ machine learning, the performance of computer vision was very limited as a lot of manual coding and developer's efforts were required. For example, facial recognition could be done using the following steps.

Create a database of individual images of all the subjects to be tracked. Store them in a specific format.

Annotate images by entering key points/features of every individual image. For example, distance between the eyes, the width of nose bridge, distance between upper-lip and nose, etc.

Capture new images and repeat the measurement process to mark the key points on the image.

All this work was done manually. Automation comes into picture with the application comparing the measurements in the new image with the ones already stored in its database. The application matches the images and tells whether it corresponded with any of the profiles it was tracking. In this scenario, there was very little automation as most of the work was still being done manually. No doubt, possibility of error was also large in this context.

Machine learning and then deep learning revolutionized the process of solving computer vision problems. Now, developers had to no longer manually code every single rule into their vision applications. Rather, programs were written to extract features to detect specific patterns in images and ultimately classify those images in different categories as illustrated in **Fig. 6.2**.

6.4 Tasks in Computer Vision

Computer vision applications recognize things in photographs by using the following techniques.

Object classification to identify the broad category of object in the photograph.

Object identification to classify the type of a given object is in the photograph.

Object verification to confirm the presence of object in the photograph.

Object localization to locate the objects in the photograph. The goal of **localization** is to find the location of a single object in an image. Usually, the located object is enclosed with a rectangular box. For example, in **Fig. 6.3**, the cat is first classified and then localized. The standard way to perform localization is to define a bounding box enclosing the object in the image.

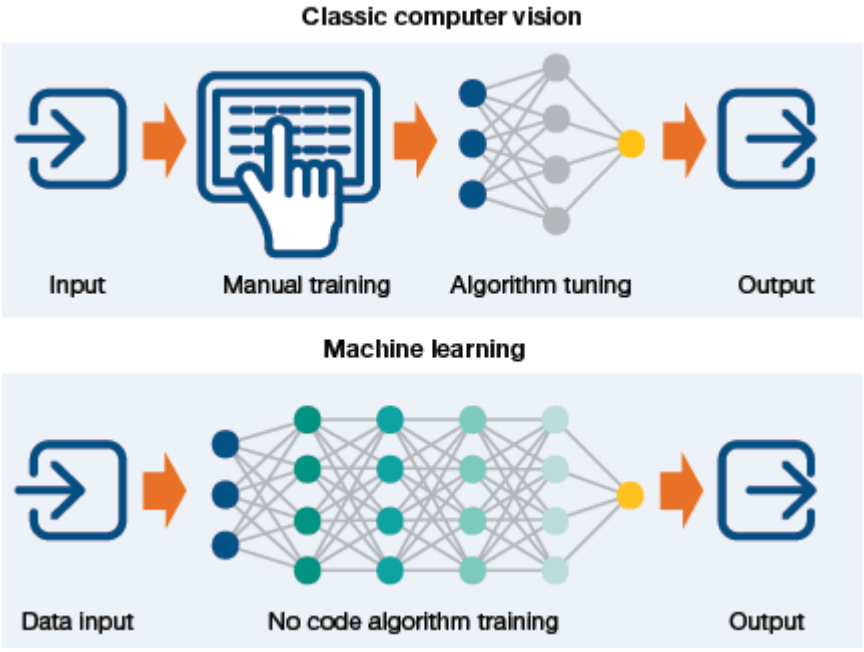


FIGURE 6.2 Classic Computer Vision vs ML Computer Vision

Object landmark detection to understand the key points for the object in the photograph.

Object detection or object identification task involves repeatedly locating and classifying all objects of interest in the image. The main aim here is to find and classify a *variable number of objects* in the given image.

Identifying unknown number of objects in a complex image is difficult for a human as some objects may be partially visible. These objects may be partly outside the frame, present in different sizes or may overlap each other. For example, object identification is extensively used to monitor real time images from security cameras to identify a specific person's face.

Object segmentation to enumerate the pixels that belong to the object in the image. **Instance or object segmentation** not only finds objects in the

image, but also creates a mask for each detected object that is as accurate as possible. It is extensively used by news channels to blur the face of a victim or any person whose identity is not to be revealed.

Object tracking inputs a video to track an object that is in motion over time. This is essentially useful in robotics (e.g., to stop a ball when a robot is used as a goalkeeper), in autonomous vehicles for spatial reasoning and path planning and in human tracking systems.



Credit: metamorworks / Shutterstock

FIGURE 6.3 Computer vision tasks

Object tracking applies object detection to capture the object being tracked and then discerns the movements of that object.

In addition, computer vision also includes methods for the following functions:

Video motion analysis to estimate the velocity of objects in a video, or the camera itself.

Scene reconstruction to create a 3D model of a scene inputted through images or video.

Image restoration uses machine learning-based filters to remove noise (to remove parts of image that are blurred).

To better realize the application of computer vision, play the game **Emoji Scavenger Hunt.**

6.5 How Computer Vision Works with Deep Learning?

With deep learning and CNN, the entire process of computer vision has been transformed. The following steps outline a general approach to building a computer vision model using CNNs.

Create a dataset of annotated images or use an existing one. Annotations include image classification, object detection (using bounding boxes), a pixel-wise instance segmentation of each object of interest present in the image.

From each image, **extract features** that are important for the current task. For example, extract facial features to recognize faces.

Train a deep learning model based on extracted features.

Evaluate the model using images that were not used in the training phase and compute the accuracy of the model.

This strategy comes under **supervised machine learning technique** as it requires a dataset from which the model can learn.

6.6 Applications of Computer Vision

Once the images are understood, computer vision is then used in the following areas.

1. **Facial recognition** systems to recognize faces in images and videos. Applications like Google photos, Snap chat, Facebook, Instagram, etc. use facial recognition system to identify people.
2. Almost every smart phone these days use facial-recognition algorithms (refer **Fig. 6.4**) to unlock phones. Facebook uses facial recognition to identify users in the posted pictures. In China, this technology is integrated with **payment technology** to allow users to do cashless

payment.



Credit: Fractal Pictures / Shutterstock

FIGURE 6.4 Facial-recognition algorithms

3. Some government agencies in different countries are also using facial technology for **surveillance**. Though the use has raised concerns among rights and privacy advocates, nonetheless, facial recognition techniques applications are on a constant rise.
4. **Content-based image retrieval systems** identify images by identifying image properties like composition, colour, texture, etc. Search engines like Google and Bing and software analysing CT scans and MRIs in hospitals, etc. used CBIR systems.
5. Computer vision is extensively used in smart interactions to give input to computers. This is especially more helpful in case of **gaming software** and in systems designed for differently-abled individuals.
6. Software used in home **security systems**, office security systems, drone-based surveillance systems, etc. (refer **Fig. 6.5**) also use computer vision techniques.
7. Computer vision is the heart of **augmented reality apps** that detects physical objects in real-time to place virtual objects within the physical environment.
8. Many agricultural organizations use computer vision techniques to monitor harvest and solve common agricultural problems including emergence of weeds or nutrient deficiency. Images collected from satellites, drones or planes are processed by computer vision algorithms to **detect potential problems** in the early phase, thereby

avoiding unnecessary losses at a later stage.

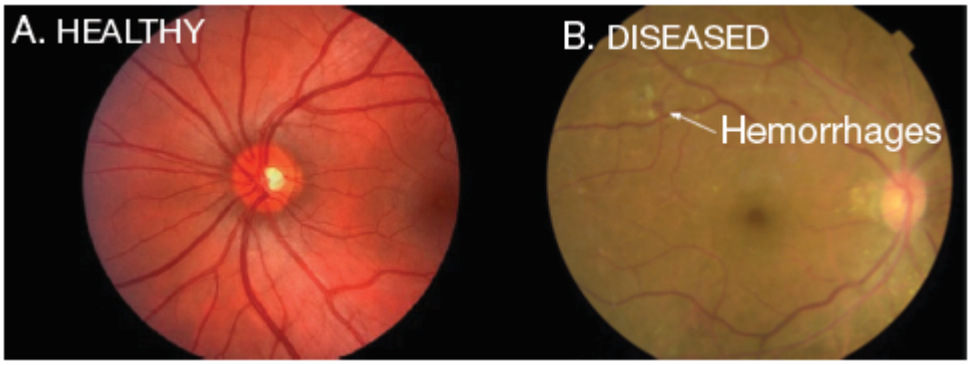


Credit: Robert Mandel / 123RF

FIGURE 6.5 Drones for Surveillance

9. **Self-driving cars (like Tesla)** use computer vision to learn from their surroundings. Every smart vehicle has cameras to capture videos from different angles. These videos act as an input to the computer vision software that processes them to detect objects like road marking, pedestrians or other cars, traffic lights, etc.
10. Websites like Facebook perform **content moderation** using computer vision techniques. Billions of posts posted by users every day are reviewed and any images/ videos that contain violence, extremism or pornography are immediately removed.
11. 90% of medical data is present in the form of images. So, most of the diseases can be easily diagnosed by processing images available in the form of **X-rays, MRI, mammography**, etc. For example, computer vision algorithms can detect diabetic retinopathy, which is the most prevalent cause of blindness (refer **Fig. 6.6**). Going further, these algorithms can also process pictures of the back of the eye and rate their severity (in case the disease is detected). Computer vision algorithms can even detect cancer with much higher precision than human doc-

tors.



*Credit (**Fig 6.6(a)**)*: Timothy Mainiero / 123RF

FIGURE 6.6 Healthy Eye vs Diseased Eye

12. **CV In Augmented Reality (AR) and Mixed Reality (MR):** In AR and MR, computer vision enables computing devices (like smartphones, tablets and smart glasses) to overlay and embed virtual objects on real world imagery. For example, CV algorithms can help AR applications detect planes such as tabletops, walls and floors. Moreover, Snapchat Filters that overlay digital images onto your face is an application of AR. Another example of AR is Pokemon Go where users can walk around with their mobile phones and find Pokemon that are overlaid on the environment around the user.
13. **Inventory management:** CV algorithms can analyse images from security cameras to generate a **very accurate estimate of the items available in the store.** Results obtained through this analysis can then be used by store managers to detect when there is an unusual increase in demand to react early and efficiently.
14. **Manufacturing:** In this sector, most losses occur due to problems in machines or because of production of defective components. Computer vision algorithms can be used for **predictive maintenance**. Images obtained through cameras (probably fitted in the eyes of robots) can be analysed to detect potential problems they occur. CV can even be helpful to reduce defects. The system can identify defects in components throughout the entire production line to notify the manufacturers to take action in real time.
15. The system is trained to classify a defect as a serious defect or a mild defect. In case of a serious defect, further production is completely halted until the issue is resolved.

16. **Insurance:** CV is used in the insurance sector for processing claims. Clients can be asked to share images while documenting and claiming money. The CV system can analyse those images to estimate and adjust repair costs, determine if the insurance covers them and even check for possible fraud. In this way, clients get a better experience as the length of claim-cycle gets shortened due to automation.
17. A CV system can also be used to **avoid accidents** by preventing collisions when integrated into industrial machinery, cars, and drones. This provision will have a huge impact on the insurance industry.
18. **Defense and security:** CV systems are used in homes and companies with high security requirements (like banks) to **identify customers** based on analysing images from security cameras.
19. Such systems can also be used to improve cargo inspection at ports or for surveillance of sensitive places such as embassies, power plants, hospitals, railroads, and stadiums. In these places, computer vision algorithm analyses and classifies images, create a detailed description of a scene in real time to help decision-makers take right action at the right time. In defense, CV systems can also be used to identify enemy terrain or enemies in images.
20. **Visual search engines:** A **visual search engine retrieves images that meets the specified criteria.** For example, Google search engine allow users to present a source image so that it can find all images similar to it. Then, applications like **Google Lens**, can be used to obtain detailed information from images. For example, you can show an image of a dog to get information about its breed, or any other details.
21. In the e-commerce market, **Pinterest Lens** can be used to find new ways to wear the dresses. For example, when you present a photo of an item, Pinterest Lens will return outfit ideas that include compatible clothing items that you would like to buy.

Amazon Go Store uses Computer Vision technique to allow customers who have installed the Amazon Go application on their mobile devices to no longer stand in long queues in the super market. They just go to the store, pick up their items and come back without paying at the kiosks.

6.7 Which Vision Succeeded?

It is interesting to note that certain experiments were carried out in Germany to address the void between computer and human vision.

Experiment 1 was conducted to understand how neural networks perceive contours (or boundary). For this, both the human and the AI participant were shown an image and asked whether it has closed contour or not. The experiment was specifically done to check if deep learning algorithm can identify closed and open shapes or not.

Results showed that a well-trained neural network could identify a closed contour even when the contour was not made of straight lines.

Experiment 2 was performed to understand whether the deep learning algorithm could identify the relation between different shapes in the picture or not. While human vision could easily understand this, it was found that deep learning algorithms that were trained with less found it difficult to draw a conclusion.

Experiment 3 was specifically carried out to identify the recognition gap that strains computer vision technology to detect an object or a thing from closer proximity when it is zoomed. When we zoom in an image beyond a certain range, we are less likely to understand what it is. However, deep learning performed better than humans and could identify minuscule features that are imperceptible to the human eye and can be detected only when the image is zoomed in very closely.

6.8 Challenges in Computer Vision

Though computer vision has emerged as one of the top fields of machine learning, there are several obstacles that makes it difficult to become a leading technology. Human vision is a complicated and highly effective system which is difficult to replicate through technology.

Given below are certain issues and challenges faced by computer vision algorithms. These are related to the nature of the data or the application

to be created and its context. In this section, we will discuss these challenges to make the CV algorithms more robust and efficient.

1. **Real-time processing:** Many applications demand the computer system to analyse images and quickly take action without any time delay. Such systems need to be very accurate. Achieving this level of accuracy with specified time constraints is not easy to achieve.

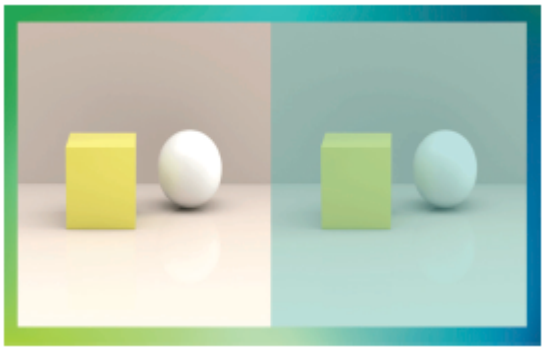


FIGURE 6.7 Color of the object changes when viewed in a different light

2. **Limited resources:** To successfully implement a computer vision algorithm, the computer system needs ample amount of memory, CPU and other computing resources.
3. **Different lighting:** Computer vision algorithms collect knowledge about the real-world objects in different kinds of lighting. For example, a white object in red light may look red in the image. Correspondingly, a red object under a red lamp becomes almost invisible (refer **Fig. 6.7**).
4. **Noise:** A noisy image or an image with incomplete/incorrect data makes it hard for computer vision to recognize objects. For example, when an image appears brighter or darker than they should be. Another example of noisy data could be images obtained from videocams on the road are much less effective when it is raining or snowing outside.
5. **Unfamiliar angles:** An image must be fed in the system from several angles. Otherwise, a computer won't be able to recognize it if the angle

changes (refer **Fig. 6.8**).



Credit: Susan Schmitz / Shutterstock

FIGURE 6.8 Object viewed in different angles

6. **Overlapping:** When there are multiple objects on the image, they may overlap with each other and remain hidden, thereby making it difficult by machine to recognize them.
7. **Different types of objects:** Objects belonging to the same category may look totally different as shown in **Fig. 6.9**. This makes it very difficult for machine to successfully recognize them.
8. **Fake similarity:** Objects looking the same may actually belong to different categories. For example, samoyed puppies look similar to little polar bears in some pictures.



Credit: creativesunday / Shutterstock

FIGURE 6.9 Same object that looks different

9. **Adversarial Attacks**: An attacker can attack a machine learning model, to make the machine fail intentionally. These flawed models are difficult to identify and may affect the system adversely.

6.9 Understanding Image Pixels

Pictures displayed on a digital screen like a monitor or are simply in digital form are made up of thousands of tiny dots also known as pixels.

The word ‘pixel’ means ‘picture element’ indicating one single dot. In coloured images, pixels of different colors are arranged alongside each other to represent different shades of colors. This means that every digi-

tized image is broken up into tiny blocks with a description of a color value at each point. **Figure 6.10** shows a pixelated black and white image of an army officer.

Future of CV: Computer vision applications are growing at a very fast pace. Corporate giants like Google, Facebook, Apple and Microsoft are investing huge amounts in computer vision.

In a coloured image, each square, or pixel stores the value of just one of the three prime colours of red, green and blue (RGB).

As seen in **Fig. 6.10**, an image viewed on a screen is made up of pixels. Each pixel comprises three strips of red, green and blue. The intensity of these strips is changed to represent a colour. To be more specific, each pixel contains a series of numbers or digits, hence the term 'digital'. These numbers dictate the colour values seen. However, when we see an image, we do not see these digits, we just see the colour that they represent.

Like RGB, CMYK is an image format that stores four values for each pixel. The pixel stores values of cyan, magenta, yellow, black.

Figure 6.10 shows the digits stored in each pixel in a black and white image as well as in a coloured image. The black-and-white image stores a '0' for a black pixel and a '1' for white pixel. This is known as a 'bitmap' image as it only requires one bit of information, 0 or 1, to describe the colour. But such a representation can be used with only two tones: black and white. Not even shades of grey can be represented using a bitmap. To represent more colours (grey shades), we usually use greyscale images that use a number from 0–255 to indicate the specific colour. A greyscale image can represent total 256 different colours.

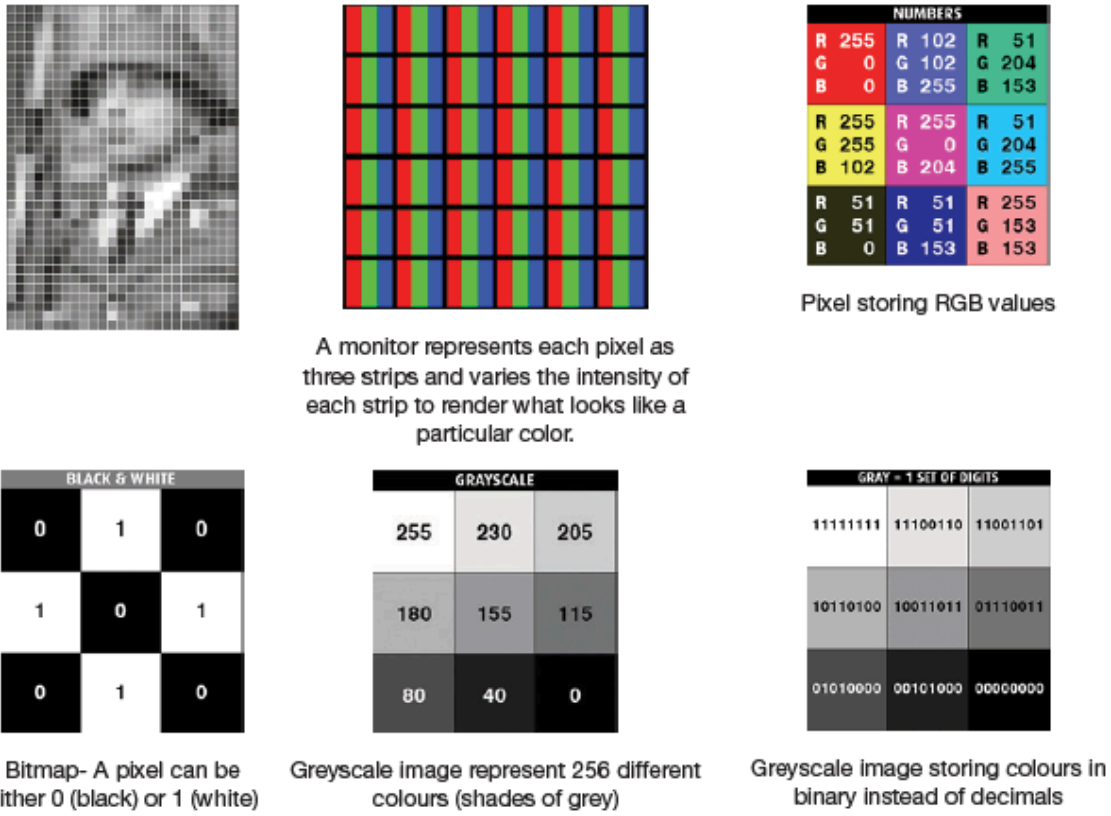


FIGURE 6.10 An Image is a composed of pixels

6.9.1 What is Resolution?

Resolution refers to the detail that an image holds. Higher the resolution, the more details the image stores and vice versa. More details mean better picture clarity. Correspondingly, less details mean less clarity. Image resolution is often measured in Dots Per Inch (DPI) and Pixels Per Inch (PPI).

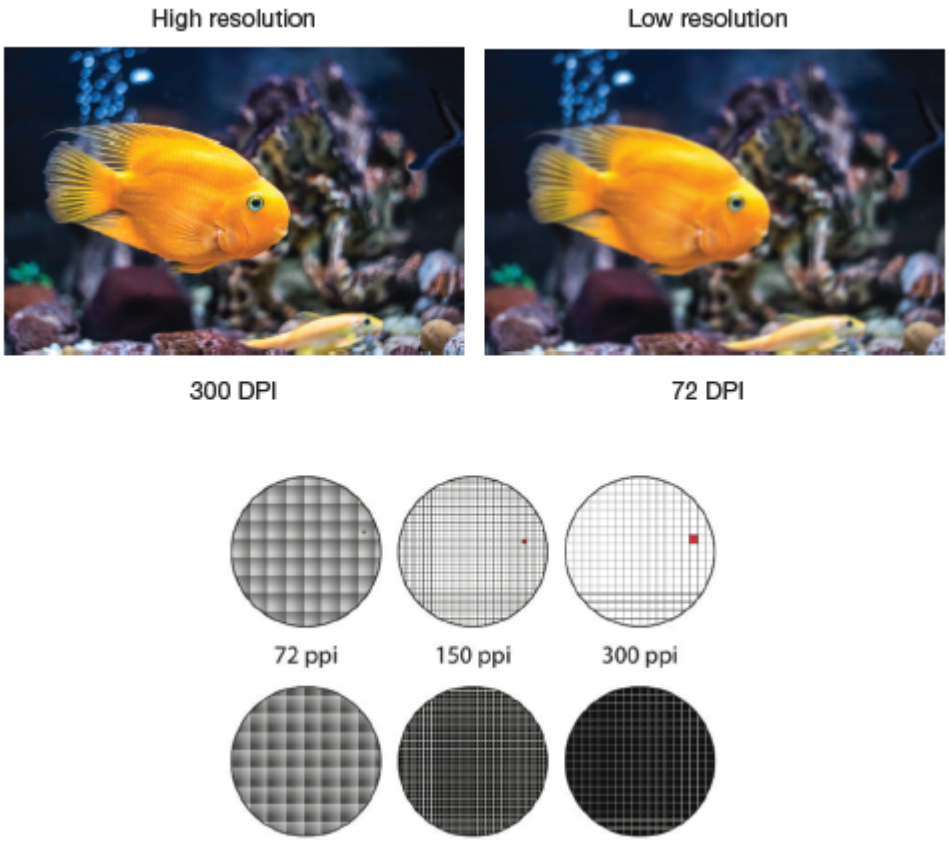
There are two conventions of specifying resolution. The first convention express resolution as the number of pixels in the width (a horizontal row) X number of pixels in height (a vertical column) of the screen. That is, if we say the resolution of the **computer monitor is 1280 X 1024**, there are 1280 pixels in a row from left-to-right and 1024 pixels from top to bottom.

The second convention expresses the number of pixels as a single number, like a **5 mega pixel camera** which means a camera having 5 million pixels (mega means millions). In our previous example, we had 1280 X 1024 = 1,310,720, or 1.31 megapixels.

Don't get confused between image size and image resolution. Image s**ize** is the dimension specified in length and width of an image. It is usually measured in pixels (px). But image **resolution**, or dpi, is the number of dots per square inch of an image when it is printed. While image **size** specifies how large the image will be viewed on monitors, **resolution** reflects printer quality.

6.9.2 DPI and PPI

DPI refers to the number of dots per inch on a printed image, whereas PPI refers to number of pixels per inch in a digital image. So, technically, it's PPI in a soft copy and DPI in a hardcopy of the image. A 300 PPI image will also be a 300 DPI image. Both the units refer to the number of units within a square inch of an image.



Credit: ZinaidaSopina / Shutterstock

FIGURE 6.11 Quality of an image depends on DPI

Figure 6.11 clarifies that we see a pixel in greater detail when we have a greater number of pixels in an inch.

6.10 Convolutional Neural Networks (CNN)

Convolutional neural networks are a special type of neural network that roughly imitates human vision. Initially, they were used to recognize handwritten digits and were especially used in the postal sectors to read zip codes, pin codes, etc. as shown in **Fig. 6.12**. A CNN requires a large amount of data and computing resources to train.

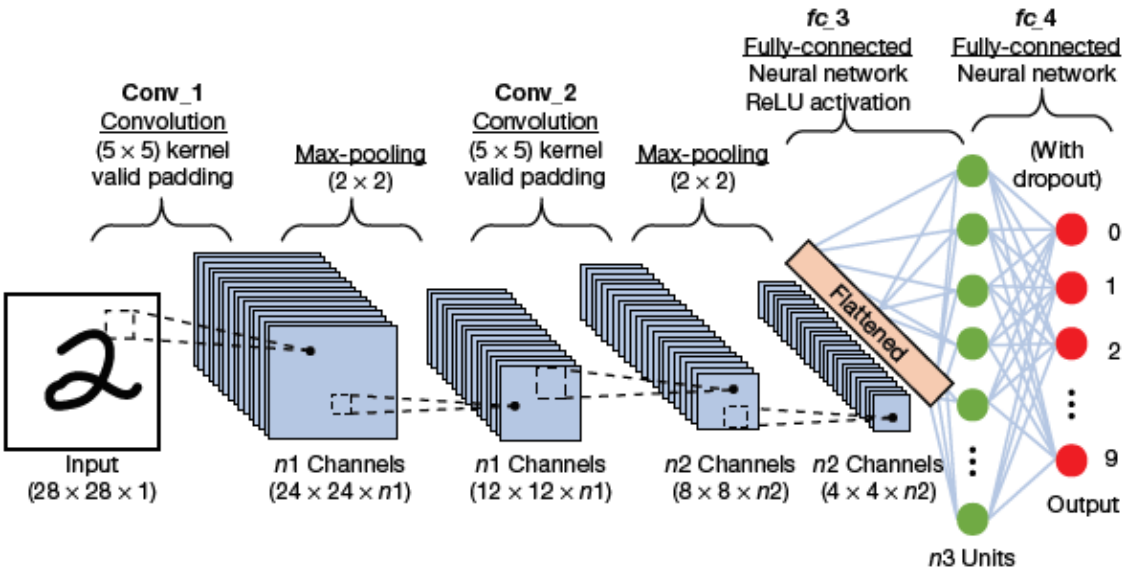
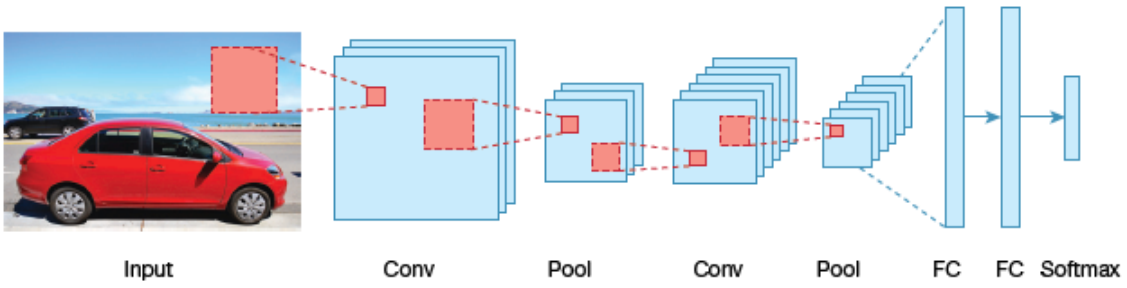


FIGURE 6.12 Classic computer vision

CNN is a branch of deep learning that uses multi-layered neural networks to analyse visual imagery. For this, CNN uses a special technique called convolution. Convolution is a mathematical operation on two functions that produces a third function that expresses how the shape of one is modified by the other.



Credit: NERAMIT SISA / Shutterstock

FIGURE 6.13 Layers in CNN

We will not discuss the mathematics part required to understand how a CNN works. Just understand that a CNN is a composition of three main types of layers (refer **Fig. 6.13**).

Convolution layer, which is the primary building block and performs computational tasks based on convolution function.

Pooling layer, that is arranged next to convolution layer and is used to reduce the size of inputs by removing unnecessary information to speedup computations *without losing features that are critical for getting a good prediction.*

Fully connected layer, arranged next to series of convolution and pooling layer to classify input into various categories.

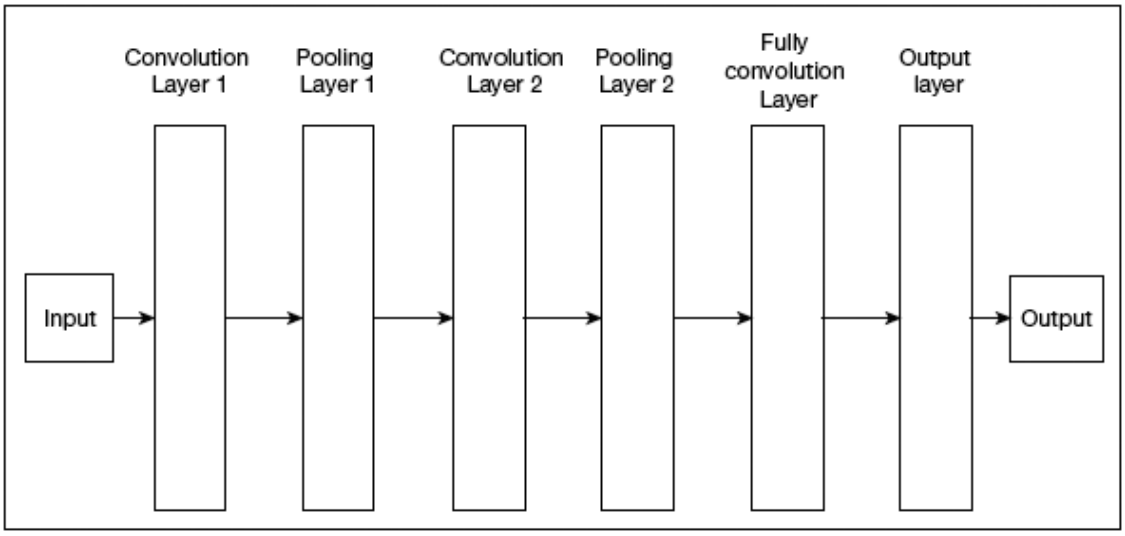


FIGURE 6.14 Convolution neural network

In **Fig. 6.14**, there are two series of convolution and pooling layer. The convolution layer receives and process the input to produce the output.

6.10.1 Understanding the Working of a Simple CNN

Convolution is a simple mathematical operation that can be applied on image processing operators. Convolution can be done by multiplying two arrays (or lists) of numbers, to produce a third array (or list) of numbers of the same dimensionality.

An image **convolution is an element-wise multiplication of image arrays and another array called the kernel followed by sum.**

For simplicity, let us consider a grayscale image that is represented as a pixel of values.

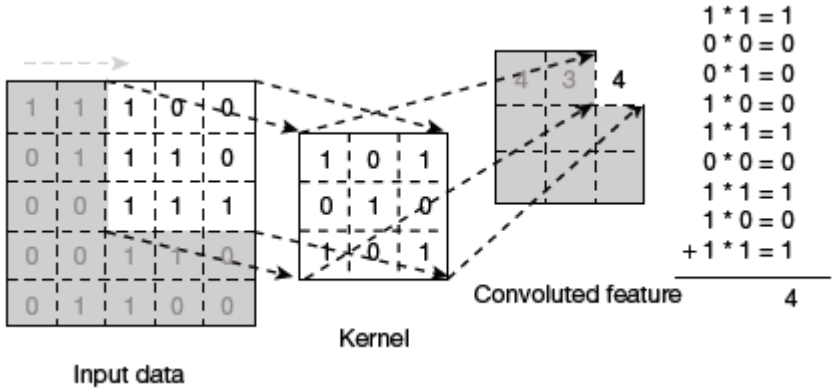


FIGURE 6.15 Kernel Layer for Filtering or Manipulating Images

In **Fig. 6.15**, we have taken a filter by using a kernel layer comprising of 3×3 matrix. **The kernel layer is applied to the input image to get the convolved feature.** This convolved feature is passed on to the next layer.

Thus, we see that a CNN consists of multiple layers of artificial neurons that calculates weighted sum of multiple inputs in order to output an activation value. When an image is input in a CNN, each layer generates several activation functions that are then passed on to the next layer. For example, the first layer extracts basic features such as horizontal or diagonal edges. This output is passed on to the next layer which detects more complex features such as corners or combinational edges. More layers in the network can be used to identify even more complex features such as objects, faces, etc.

Activation map (also known as feature map) generated by the final convolution layer guides the classification layer to produce a set of confidence scores (values between 0 and 1) indicating how likely the image is to belong to a particular class.

Therefore, a **CNN need not be limited to only one convolutional layer**. The first convolution layer captures low-level features such as edges, color, gradient orientation, etc. Subsequent layers can extract other high-lev-

el features that gives a wholesome understanding of images in the dataset. In the convolution layer, **several kernels are used to produce several features.**

The activation or the feature map can then be used to

- reduce the image size so that it can be processed more efficiently.
- focus on certain features of the image that can help in processing the image further. For example, to recognize a person, we may need to focus on a person's eyes, nose and mouth to recognize the person rather than focusing on the entire face.

6.10.2 THE Pooling Layer

The convolution layer is followed by a pooling layer that can be of two types—average pooling and max pooling.

Max pooling finds the maximum value of a pixel from a part of the image covered by the kernel. It discards the noisy activations thereby performing de-noising along with dimensionality reduction.

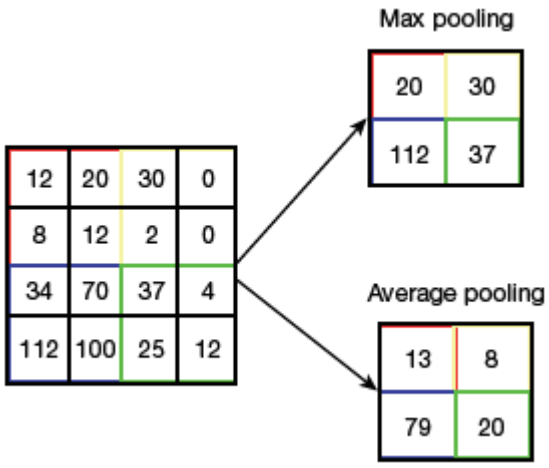


FIGURE 6.16 Pooling Techniques

Average Pooling returns the **average of all the values** from the part of the image covered by the Kernel. **Max Pooling performs better than Average Pooling**.

The two techniques of pooling can be better understod from **Fig. 6.16**.

6.10.3 Limitations

The advent of CNN has revolutionized the field of AI in a big way. A CNN gives in-depth results and recognizes even minute patterns and details that otherwise goes unnoticed to the human eye. However, even after using great amount of computing power while being trained with massive amount of data, CNN were not able to completely identify, block and remove inappropriate content from social media.

CNN also fails to detect objects when they are seen under different lighting conditions and presented from new angles. **Table 6.1** highlights the differences between a CNN and a neural network, often referred to as ANN or artificial neural network.

TABLE 6.1 Differences between CNN and ANN

CNN	ANN
• Partially connected layers	• Fully connected layers
• Can be used for image of any size	• Can be used for small images
• Considers only a part of the entire image for learning patterns (depending on filters)	• Considers the entire image for learning patterns
• Comparatively a smaller number of parameters used for learning	• Large number of parameters used for learning
• More efficient technique	• Less efficient technique
• More computational cost	• Less computational cost
• Once a CNN has learnt to detect a pattern in one position, it can detect the same pattern even in different locations	• Once an ANN has learnt to detect a pattern in one position, it cannot detect the same patter even in different locations (e.g., if the image is rotated or shifted)
• More records/data for training is required	• Number of records/data for training depends on complexity of the task
• Success rate is high	• Success rate is comparatively low.

Application Example of Image Features

In computer vision and image processing, a **feature** is a piece of information that is required to solve a computational task in context to a given application. These features may be points, edges or objects (refer **Fig.**

6.17). For example, if given six small patches of images, we can make our algorithm identify location of those image patches in the image. Once developed, we can use this algorithm to find similar images in a huge dataset of images.

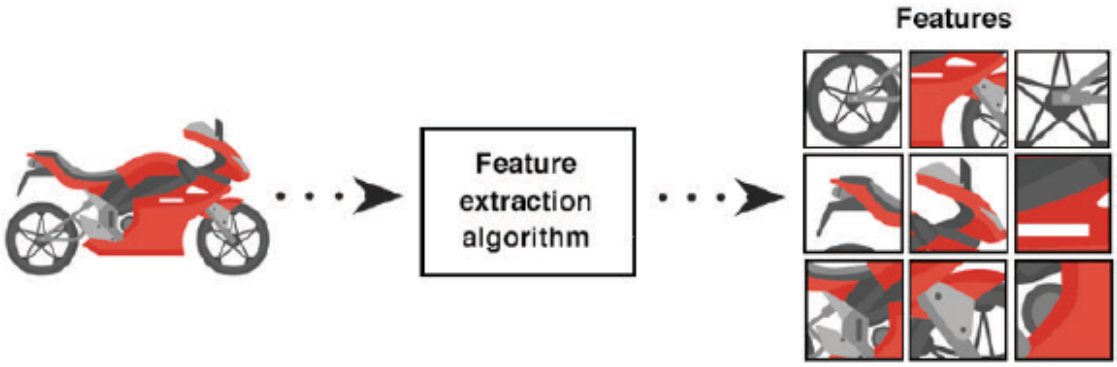


FIGURE 6.17 Extracting Features from an Image

6.11 Working with Images Using OpenCV

OpenCV is one of the most popular libraries providing functions for computer vision. The library has functions for the following tasks.

1. Reading an image
2. Extracting the RGB values of a pixel
3. Extracting the region of interest (ROI)
4. Resizing the image
5. Rotating the image
6. Drawing a rectangle
7. Displaying text

With OpenCV, users can easily do complex tasks such as identifying and recognizing faces and objects, tracking moving objects, stitching images together to generate an entire scene with a high-resolution image, etc.

To install OpenCV on windows using pip command, just write the following command in Window's Command Prompt.

```
Pip install opencv-python
```

Before using the OpenCV package, let us first discuss some important function that we will be using in this section.

The file should be in the working directory or we must give the full path to the image.

The **imread()** function accepts name of the image to open and a value 1 if the image has to be read as a coloured image. The syntax of this function can be given as follows:

```
cv2.imread(path, flag)
```

Here, the path is a string representing path of the image to be read and flag specifies how the image should be read—0 for grayscale image and 1 for a coloured image.

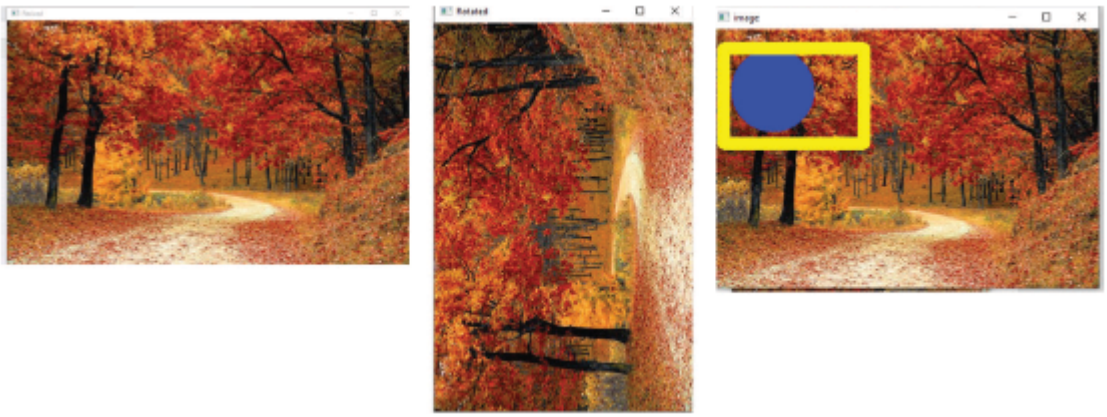
The imread() either returns the image loaded from the specified path file (if the image is found), else returns a matrix.

Now, type the following programming in Python IDLE to read and process images using OpenCV.

Example: Reading and Processing an image

```
import cv2
import numpy as np
 #importing the opencv module
img = cv2.imread('Nature.jpg',1)
print(img.shape)
# Resize image
img_resized=cv2.resize(img, (780, 540))
cv2.imshow("Resized",img_resized)
# Rotate image
image = cv2.rotate(img, cv2.ROTATE_90_COUNTERCLOCKWISE)
# Draw a circle on the image
cv2.imshow("Rotated",image)
cv2.circle(img,(80,80), 55, (255,0,0), -1)
cv2.imshow('image',img)
# Draw a rectangle on the image
cv2.rectangle(img,(15,25),(200,150),(0,255,255),15)
cv2.imshow('image',img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

```
OUTPUT
(2848, 4288, 3)
```



The **resize()** is used to change the size of the image. In ML algorithms, increase in number of pixels in an image increases the number of input nodes and thus the complexity of the model. The syntax of resize() is,

cv2.resize(s, size)

where,

s is the input image (required).

size is the desired size for the output image after resizing (required)

The **cv2.rotate()** method is used to rotate a 2D array in multiples of 90 degrees. The syntax of this method is,

cv2.rotate(src, rotateCode[, dst])

where,

src is the image to be rotated.

rotateCode specifies how to rotate the array. Its values can be cv2.ROTATE_90_CLOCKWISE, cv2.ROTATE_180, cv2.ROTATE_90_COUNTERCLOCKWISE.

The **imwrite ()** function having syntax,

cv2.imwrite(filename, image)

is used to save an image on a storage device in the specified location. The function returns a value (or status) indicating success or failure in saving the image. The code given below reads the image in file Nature.jpg and saves it in C:\ -> Users -> Chikki Folder

```
img = cv2.imread('Nature.jpg',1)
status = cv2.imwrite(r'C:\Users\Chikki\cat.png',img)
print("Image written success? : ", status)
```

If the file is successfully written then this function returns True otherwise returns False.

Basic Operation on Images

In this section, we will discuss some basic operations that can be performed on the images. These are, **Access pixel values and modify them. This can be done by writing**

```
img = cv2.imread('Nature.jpg',1)
px = img[50,60]          # accessing a pixel
print( px )

Output:
 [157 166 200]
```

This prints a list containing 3 values that in turn specifies color of the pixel- Blue component, Green component and Red component. To modify the values, can access the pixel and then overwrite it with new value as shown below:

```
img[50,60] = [250,125,55]
print( img[50,60] )

Output:
 [250 125 55]
```

Access Image properties like the size (total number of pixels in the image), number of rows, columns, and channels. While shape() is used to print the number of rows, columns and colors, size() prints the size of the image.

```
print( img.shape )
>>(350, 348, 3)
print( img.size )
>> 4248148
```

Image ROI (Region of interest) to select only a specific region of the image. For example, to detect eyes in an image, we need not search the entire image, we must concentrate only on face. Thus, here the face is our ROI.

Splitting and Merging Image Channels to split the channels from an image and then work on each channel separately or to merge the channels together.

OpenCV Drawing Functions

OpenCV has functions to allow users to draw certain shapes on an image such as circle, rectangle, ellipse, polylines, convex, etc. This is often used

to highlight any object in the input image. For example, Facebook identifies face by drawing a rectangle around it.

Drawing circle: circle() is used to draw a circle in an image. Syntax of circle() is:

```
cv2.circle(image, center_coordinates, radius, color, thickness)
```

where,

image is the input image on which a circle is to be drawn.

center_coordinates specify the X and Y coordinate values representing the center of the circle.

radius is the radius of the circle.

color specifies the color of the border line of the circle.

thickness indicates the width of the circle border line in px.

Drawing Rectangle: We use the rectangle() to draw a rectangle on the image. Syntax of rectangle() is,

```
cv2.rectangle(image, start_point, end_point, color, thickness)
```

where,

image is the input image on which rectangle is to be drawn.

start_point specifies the top left vertex X and Y coordinates of the rectangle.

end_point indicates the bottom right vertex X and Y coordinates of the rectangle.

color specifies the color of the border line of the rectangle to be drawn (in the form of BGR).

thickness gives the width of the rectangle border line in px.

Drawing Lines: The line() is used to draw lines on an image. Its syntx can be given as,

```
cv2.line(image, start_point, end_point, color, thickness)
```

where,

image is the input image on which line is to be drawn.

start_point specifies the starting coordinates (X and Y values) of the line.

end_point gives the ending coordinates (X and Y values) of the line.

Color indicates the color of the line to be drawn in the form of BGR)

thickness is the width of the line in px.

Write text on an image: Text can be written on an image by using the putText(). The syntax of this function can be given as,

```
cv2.putText(img, text, org, font,fontScale color)
```

where,

img represents the input image on which we have to write text

text is what we have to write on the image.

org denotes the Bottom-left corner of the text string on the image. Thus, org is used to set the location of text on the image

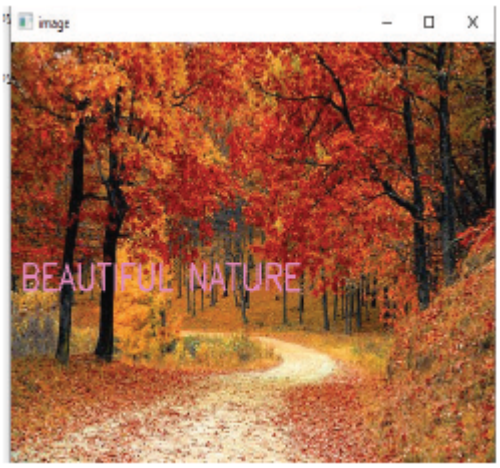
font specifies the font of text.

fontScale denotes the scale of the font by which size of the font can be increased or decreased.

color represents the color of the text in BGR form.

Example: Code given below places text on an image.

```
import cv2
import numpy as np
 #importing the opencv module
font = cv2.FONT_HERSHEY_SIMPLEX
img = cv2.imread('Nature.jpg',1)
cv2.putText(img,'BEAUTIFUL NATURE',(10,200), font, 1,(200,125,235),2)
#Display the image
cv2.imshow("image",img)
cv2.waitKey(0)
```



Canny Edge Detection: Edge detection is an image processing technique used to identify object's boundaries within an image. In the code given below, we have used the Canny Edge Detection algorithm. Syntax of Canny() can be given as,

edges = cv2.Canny(img, minVal, maxVal, apertureSize, L2gradient)

where,

img is the input image whose edges are to be detected.

minVal is the minimum intensity gradient

maxVal is the maximum intensity gradient

As per the syntax, edges with intensity gradient more than maxVal are definitely edges and those with intensity gradient less than minVal are sure to be non-edges and therefore discarded. Edges with values between maxVal and minVal are classified as edges or non-edges based on their

connectivity with the 'sure edges'. If the edges are connected to "sure-edges" then they are taken to be a part of the edges. Otherwise, they are treated as non-edges and discarded.

Example: Detecting Edges of an image.

```
#importing the opencv module
import cv2
# using imread('path') and 1 denotes read as  color image
img = cv2.imread('Nature.jpg',1)
edges = cv2.Canny(img,200,300,True)
cv2.imshow("Edge Detected Image", edges)
cv2.imshow("Original Image", img)
cv2.waitKey(0)  # waits until a key is pressed
cv2.destroyAllWindows() # destroys the window showing image
```

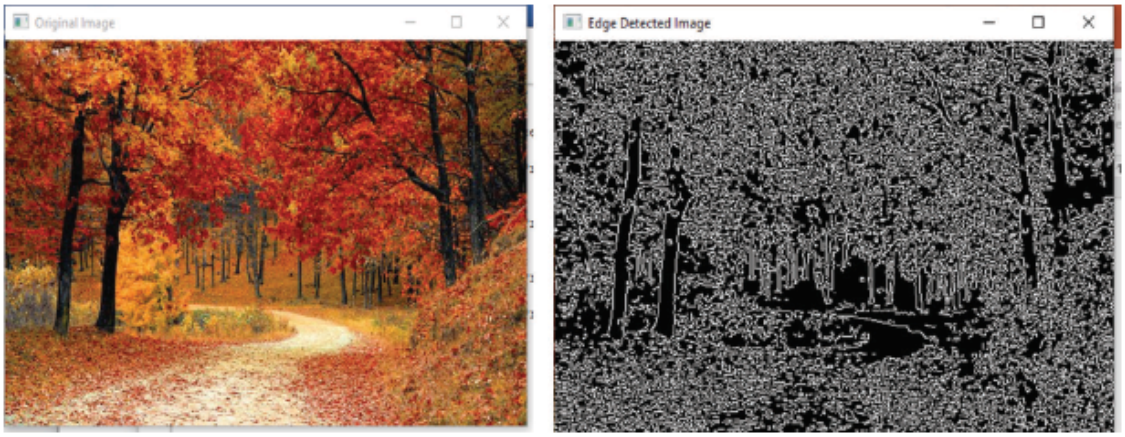


Image Smoothing is an image processing technique that removes noise from an image. For example, when blurring an image, low-intensity edges are removed. Blurring is often used to hide the details or any confidential information in an image. Different types of blurring techniques supported by OpenCV includes: averaging, median Blur, Gaussian Blur and Bilateral Filter.

In the averaging technique, the image is normalized with a box filter. Average of all the pixels in the box area is calculated. The average value then replaces the value of the pixel at the center of the box filter. The cv2.bur() in OpenCV is used to perform this operation. It's syntax can be given as,

cv2.blur(src, ksize)

where,

src specifies the image to be blurred.

Ksize represents the blurring box area.

```
#importing the opencv module
import cv2
# using imread('path') and 1 denotes read as  color image
img = cv2.imread('Nature.jpg',1)
cv2.imshow("Original Image", img)
cv2.imshow('cv2.blur output', cv2.blur(img, (3,3)))
#cv2.imshow('cv2.GaussianBlur output', cv2.GaussianBlur(img, (5, 5),
cv2.BORDER_DEFAULT))
cv2.waitKey(0)  # waits until a key is pressed
cv2.destroyAllWindows()  # destroys the window showing image
```



median Blur technique calculates the median of all the pixels under the box area and replaces the value of the central pixel with this median value. medianBlur() provided in the OpenCV pakage is used to easily implement this kind of smoothing. Syntax of this function can be given as,

cv2.medianBlur(src, dst, ksize)

where,

src represents the input image.

dst represents the output image.

ksize represents the size of the kernel (box).

```
import cv2
# using imread('path') and 1 denotes read as  color image
img = cv2.imread('Nature.jpg',1)
cv2.imshow("Original Image", img)
cv2.imshow('cv2.medianBlur output', cv2.medianBlur(img,5))
cv2.waitKey(0)  # waits until a key is pressed
cv2.destroyAllWindows()  # destroys the window showing image
```



Gaussian Blur technique, instead of the box filter, a Gaussian function is used to blur the image. The width and height of the kernel window should be specified and these values should be positive and odd. Standard deviation in X and Y coordinates are also specified using the sigmaX and sigmaY parameters respectively. If sigmaX = sigmaY = 0, then they are calculated from the kernel size. The GaussianBlur() is used to implement this technique. Its syntax is,

GaussianBlur(src, dst, ksize, sigmaX,sigmaY)

Where,

src is the image to be blurred

dst is the output image of the same size and type as src.

ksize specifies the size of the kernel.

sigmaX and **sigmaY** are double values that represents standard deviation in X direction and Y direction respectively.

```
import cv2
# using imread('path') and 1 denotes read as  color image
img = cv2.imread('Nature.jpg',1)
cv2.imshow("Original Image", img)
cv2.imshow('cv2.GaussianBlur output', cv2.GaussianBlur(img, (5, 5),
cv2.BORDER_DEFAULT))
cv2.waitKey(0)  # waits until a key is pressed
cv2.destroyAllWindows()  # destroys the window showing image
gd
```



Bilateral Filter is a highly effective technique to remove noise but is slower compared to other filters. The problem with Gaussian filter is that it blurred the edges also which was not at all desirable. Bilateral filter blurs only the pixels with similar intensities to the central pixel thereby preserving the edges (pixels on edges have large intensity variation). It is implemented using the cv.bilateralFilter().

6.12 Immersive Experience

Immersive experience is any technology that extends reality or creates a new reality by leveraging the 360 space. While some types of immersive experience extend reality by overlaying digital images on a user's environment, others create a new reality by completely immersing users in a digital environment, shutting them out from the rest of the world. This gives as more engaging or satisfying experience.

The Skyview app is an example of AR in action, allowing you to see where constellations are in the sky in real-time as you move your phone around.

The simplest tool to enjoy immersive content is through smartphones or specialized headsets. On the software side, although a web app or a desktop application can be used, web app is more popular.

To better understand the concept of virtual reality, just imagine how exciting it would be if you send your digital avatar to attend a meeting on your behalf. The avatar not only attends but also jots down the minutes, and reports to you after the session is over.

6.12.1 Elements of Immersion

An immersive experience is defined as a multi-sensory experience (sight, sound, touch and scent) across a journey, or a task that is contextually relevant to create intuitive and emotional value for the user that drives a better relationship.

Sight VR headset blocks out the peripheral vision or use a wrap-around headset to enhance vision so that the attention of the wearer is focused on what is happening directly in front of him/her. On the contrary, AR uses headsets or smartphone displays to add virtual elements to the real world.

Sound VR headsets include sound-dampening headphones that force the wearer to focus on the sounds of the virtual world. AR on the other hand, provides sounds for whatever is taking place on the screen.

Touch VR accessories provide haptic feedback for the wearer. Moreover, effects like vibrations and rumbles are used to synchronize with the virtual world. AR, however, does not need touch to increase immersion.

Thus, all these elements together create an immersive experience.

True immersion depends on a suspension of disbelief and a willingness to be transported to a different world.

6.12.2 Types of Immersive Experiences

360-degree content is an interactive photo or a video recording shot in every direction at the same time. This allows users to change the viewing direction at any moment. Digital twin is a precise virtual model of a real-life object, process or system. We can even create a digital of an object that does not even exist. Such models can be used to display textual details or to monitor, control and run simulations.



Credit: insta_photos / Shutterstock

FIGURE 6.23 Immersive Experiences

Technically, 360° content integrates two or more videos in a dome-like space. These videos are shot from all possible angles, with 2D instructional content overlaid on top of them. This technology is extensively being used to walk customers through a detailed range of product features and benefits. This helps users to get a wide-eyed version of the product in the real world. Similarly, 360° content is being used in travel and tourism industry to help tourists pre-visit a particular destination.

Virtual reality (VR) is a computer simulation that immerses the user inside a virtual world and allows him to interact with the environment. VR stimulates user's senses to make him feel as if he was actually there in that environment.

PwC predicts that VR and AR could add $1.8 trillion to the global economy by 2030.

VR shuts a user completely from the rest of the world while being surrounded by content. Through a head mounted display (HMD), whatever content the user experiences in the headset becomes the reality for him.

360 vs 360 VR: Though the two terms are same, 360 VR is a combination of 360 content with VR mode (refer **Fig. 6.23**). It allows users to view the 360 contents in a cardboard headset, or sometimes even a mobile 360 VR headset. Users can experience 360 contents without the help of an HMD and is best experienced on mobile.

Mobile VR leverages HMDs that are connected to smartphones. Examples of these are the Samsung Gear VR,_ Google's Daydream or even Google Cardboard.

True VR uses headsets with powerful computers or consoles. Examples include HTC Vive, Oculus Rift, Playstation VR, etc. These powerful HMDs use sensors (separate from the HMD) to keep a track of a person's movement and his surroundings. The sensors would then adjust the VR content to whatever is within a user's environment. True VR uses CGI and 3D modelling for content.

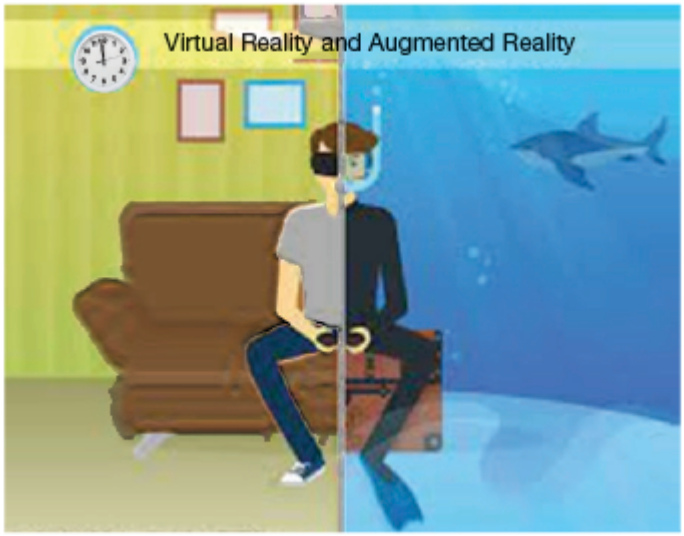


FIGURE 6.24 AR vs VR

360 VR vs In-VR: With 360 VR, users can use their device to explore content by looking in any direction but not change the content that surrounds the objects. In-VR content takes depth into consideration. This means that when a person is viewing an object, he can move closer to the

object or move further away from it. The entire content is adjusted accordingly.

Augmented reality (AR) enhances the user's perception of the real world by adding a computer-simulated layer of information on top of it (refer **Fig. 6.24**). While AR adds or hides data from the environment, VR completely replaces the user's perception of the real world.

Using AR, digital images are presented on top of the real world. This means that users who leverage AR are not completely shut off from the world. Instead, AR extends their reality.

For example, Snapchat Filters that overlay digital images onto your face is an application of AR. Another example of AR is Pokemon Go where users can walk around with their mobile phones and find Pokemons that are overlaid on the environment around the user.

Many stores like Target and Ikea have their own AR apps where users can choose a product and place it virtually on anything to see how it looks. For example, you can choose a dress and see how it looks on you. This allows consumers to 'try out' a product before even buying it.

AR also provides the ability to 'touch' the product. Some examples are as follows:

1. BMW gives its users a chance to try out 3D models of their cars.
2. Burger King used the motto 'Burn That Ad' in its advertising campaign in Brazil. In this campaign, the user is asked to scan any competitor's ad and 'burn' it by using augmented reality. The most active users are then rewarded with a free branded whopper.

TABLE 6.2 Differences between augmented reality and virtual reality

Augmented Reality	Virtual Reality
Overlays computer-generated 3D content in the real world.	Visually immerses users with simulated objects and environment.
User is connected with both physical as well virtual world.	Completely disconnects users from the real world. The user is totally engrossed in the virtual world.
It is experienced using smartphones, tablets or AR wearables.	It is experienced using virtual wearables.
It is usually used for demonstrations, designing and mapping.	It is mainly used in medicine, military, gaming, etc.
Users can rotate, move, scale and manipulate objects in the real world.	Users can rotate, move, scale and manipulate objects in the virtual world.
Users can interact in the real world while using an AR application.	Users are cut-off from real world for any interaction.
Users can physically move in their environment.	Users move only virtually in the environment.
Users maintain a sense of presence in the real world.	Entire senses of users are under the control of the virtual environment.
Users feel and know that they are a part of the real world.	Users are shut-off from the real world.

Mixed Reality (MR) brings together VR and AR into an enhanced version of AR. Mixed reality is also referred to as AR 2.0 because it achieves better immersion than AR. MR integrates virtual objects in the real world. It responds to changes in the environment and to user interaction in real-time.

Unlike VR, MR does not shut out users from the rest of the world. Instead, the Head Mounted Display (HMD) is more like a pair of glasses that overlays digital images on top of your environment (just like AR). Mixed reality is extensively being used for commercial, developmental and entertainment purposes, for the following:

1. Conducting simulation-based training especially military training without increased risks.
2. Creating an interactive environment with the full inclusion of virtual objects in reality.

For example, Disney World uses a combination of AR/MR to create a game to entertain park visitors while they walk around and wait in line for physical attractions.

Extended reality (ER) is used to enhance or completely replace reality. It is an umbrella term for every immersive technology including augmented reality, virtual reality, mixed reality, digital twins, etc. This technology extends reality by adding digital counterparts in the form of inscriptions, videos or animations.

ER is usually used for training employees, giving a walk through a live hologram of a product to understand its features.

Digital Twins: They are near-exact virtual models of real-life objects, processes or systems. This technology is usually used in manufacturing or engineering to simulate physical things either for optimizing or for studying how they behave before actually building them.

Accenture uses full-scale XR to create highly personalized experiences for its clients through sight, touch and sound.

For example, NASA uses digital twins to monitor and optimize satellites in space from the ground. Digital twins of satellites help scientists to foresee potential dangers, fuel shortage, engine power, steering power and other parameters of any spacecraft mission.

Mercedes uses this technology to optimize the performance of its F1 cars. Digital twin is also extensively used in healthcare and for digitally recreating entire cities. Therefore, a **digital twin** represents a ‘carbon-copy’ of a real-life component, structure or process which can be easily tracked in real-time. Digital twin technology combined with 3D-modelling techniques helps users to actively interact with the environment.

Advantages

1. Digital twins are used for quality assurance tasks.
2. It provides quick insights into a product anytime, anywhere for faster decision making.
3. It helps to visualize defects in the real-time performance of certain equipment.

4. It troubleshoots machines at remote locations.

6.12.3 Applications of Immersive Experiences

Immersive technology is not a new term for those who read Sci-Fi books or watch in Sci-Fi movies.

Business: Every business generates a number of reports using ERP software. These reports are, at times, difficult to understand. In such a case, an AR headset is used which will see only the most important bits of data in the report. The headset will show all the details when the user looks at specific parts of the report.

Using a VR headset can make remote business meetings through tele or video conferencing more authentic.

E-commerce uses 360-degree technologies to provide better visualization. It allows users to look at the product from every possible angle. Many websites have made the 360-degree content even more interactive. It can react to changes in options in real time, giving the user an immediate preview.

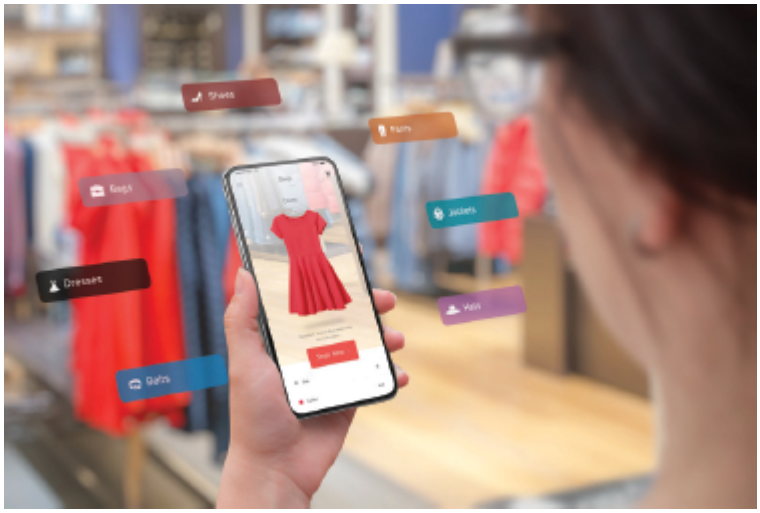
These websites are even using MR to show how the product will look like in a real-life environment.

Architecture and design are indeed extensively using immersive technology. With VR, we can explore building designs in a 1:1 scale. AR is used to show additional relevant information. MR allows users to edit 3D models created by them as per user's requirements. Any small changes and adjustments in the simulation are reflected instantly.

Engineering, maintenance and repair of complex equipment are easily done using immersive techniques. While repairing, engineers use AR headsets to get useful data right away. Artificial intelligence can help them by giving suggestions about the next best action that can be taken.

In the design phase, a digital twin is used by engineers to see their creations in a virtual environment. They no longer need to build prototypes.

Similarly, during testing phase, a digital twin can be used for running simulations and collecting analytical data.



Credit: Stanisic Vladimir / Shutterstock

FIGURE 6.25 AR for Shopping

Healthcare is another area that extensively uses immersive experiences. Augmented and virtual reality both can make the current state of medical training more effective. While AR provides useful information to healthcare employees when learning how to use new equipment, VR can help them to practice more complex medical procedures in a virtual environment before actually performing them on a live patient.

Similarly, in radiology, AR is used during pathological tests to study medical procedures and evaluations. By virtually mapping the X-ray data of a patient, radiologists can identify abnormalities of bones and classify the disease more quickly than before.

Telemedicine uses VR or MR to break the distance barriers between patients and practitioners. Patients can now get themselves diagnosed in a more comfortable setting and having more accessible medical services. An AR headset can be used to give important information to every patient in a hospital, keeping their mind busy and divert it from all worries.

Even VR solutions are used to reduce anxiety in patients, helping them with sleep or just serve as a positive distraction. AR can be used to visualize patient data and treatment assistance. An AR headset can show only

the relevant information directly on the patient's body or give suggestions regarding the medicine to be prescribed. Moreover, during surgeries, immersive technologies can be used to outline the most optimal cutting path, give warnings about potential problems and help with additional information.

The travel industry is using immersive technologies for selling real-life experiences. For example, 360-degree content and VR provide customers with the option to try before they buy something. Customers can have a virtual tour of the place whether they intend to go, virtually walk around the hotel they intend to book. This will ensure them that they are making the right choice. AR headset can serve as a 24 × 7 personal tour guide that provides additional information in real time, making sightseeing easier and more informative, finding cafes, restaurants, hotels, inns and landmarks.

Real estate uses immersive technologies to virtually show the property that has to be taken on rent or for sale. It is very inconvenient and time consuming for the buyer as well as the seller to see every property. Buyers can check out properties remotely whenever they have time; that too without the help of a real estate agent. Later, they can personally visit only selected properties and use an MR headset to get information about the property, decoration options or furnishing suggestions.

DHL could cut 25% cost by using AR in their logistic business. Automaker Volvo uses VR headsets in the testing process and BMW uses it for workstation planning.

Augmented reality in logistic centres uses immersive technology to reduce warehouse costs by using AR smart glasses in the picking process. The smart glasses show a list of stuff to be picked (reducing lookup times) and calculate an optimal route (to lower travel time). Headset are used to read barcodes and determines if the current location is the right one. In case the location is not correct, the headset guides the workers to the correct location.

Automakers are using MR for testing automobiles in simulated extreme situations. Sensors in headsets enable precise measurement of reaction

times and other important safety factors. VR is also used in workstation planning to reduce the labour force required and assess processes completely virtually. AR headsets can be used by employees for engine assembly units training at a faster pace (approximately three times faster).

While teaching about planet Mars, teachers can use MR to give a virtual tour to mars to explore its surface.

The car industry is also using immersive technology to create virtual showrooms where consumers can look at cars, take virtual test drives and even virtually sit in actual car seats.

At **McLaren**, VR software is used to design car models. A live hologram of the car is made which designers can alter. Even Volvo engineers use VR software to simulate drive trials and customize designs accordingly.

Companies like Adidas use VR to enhance workflow by using VR as a virtual team collaboration space. They use it when different teams have to work together. VR gives employees the ability to be in the same virtual room and share 3D models and other content.

Walmart is using VR headsets for training employees. They use immersive experience to prepare their employees for different real-life situations.

Education: VR can help students to learn about a particular concept. For example, a VR headset can be used to take a virtual tour of a historical monument and learn about historical events of importance associated with it. Similarly, MR can be used to learn about human body and how each part functions by overlaying digital images on an actual person and highlighting where the different parts are located.

Gaming is among the first applications that extensively used immersive technology. The launch of Playstation VR, Pokemon Go and VR arcades are examples of application of this technology. **Table 6.3** list some games and the underlying technology they use.

TABLE 6.3 Games using Immersive Experiences

Game	Technology	Description
Cosmos VR	VR	An audio-guided journey through the universe that facilitates users to visit many of its corners and galaxies.
Minos Starfighter VR (End Space VR in the App Store)	VR	An arcade space shooter with great design and tempting blasters that provides support for a Bluetooth joystick.
Bad Dream VR	VR	A horror game, where the user wakes up in an unfamiliar place and tries to escape from those who are following him/her.
Pokémon GO	AR	A game in which the user walks in real terrain and uses the smartphone's camera to catch Pokémon that appears on the screen when the user reaches the desired point.
Jurassic World Alive	AR	An educational game with a dinosaur raising simulator, and mini-games. The user explores the real world by using geolocation and searches for the DNA of prehistoric monsters. Having collected enough elements, the player creates a pet in the laboratory, and then trains it to fight other creatures in PvP battles.
Harry Potter: Wizards Unite	AR	Game with unique magic features. Each new player becomes an employee of a special unit of the Ministry of Magic and fights the disasters that have befallen the world of wizards.
Ghosthunters: Slimer AR	AR	A game of hunting ghosts and clearing their city of evil creatures.

Immersive storytelling and virtual try-on: AR places users in the centre of the story, captures their attention or showcases products right on their faces. For example, cosmetic brands like L'Oreal and Maybelline allow consumers to apply virtual makeup to boost sales among Gen Z. Similarly, Turkish Airlines has created a branded filter on Instagram called 'Turkish Airlines My Dream Destination Project' that immerses people into virtual travels during the pandemic.

Designers combine traditional 2D CAD applications with VR tools to perfect their designs. Examples include softwares like V-Ray or SketchUp.

Aviation Management: Immersive technology allows flight engineers and ground handlers to identify, probe and analyse the situation before the aircraft takes off. This helps the crew to perform aircraft maintenance and engineering checks from a remote location. AR/VR headsets put teams on airside so that they can interact with aircraft holograms to conduct visual inspections or cargo inspections.

Supply chain and logistics: VR reduces logistic costs by providing virtual-navigation routes for the pickup of goods. This not only increases efficiency of the employees but also expedites the logistics process (refer **Fig. 6.26**).

MR devices support professionals working in supply chain by superimposing routes over truck windshields. Similarly, AR devices help by notifying teams when supplies perish.



Credit: Zapp2Photo / Shutterstock

FIGURE 6.26 Using Immersive Experiences for Handling Logistics

Entertainment sector: These days, especially after the outbreak of COVID-19, many events are conducted virtually. Immersive technology compensated for the lack of physical presence. Immersive devices help users to experience a concert or movie as if he/she are a part of it (refer **Fig. 6.27**).

Commercial businesses using immersive experience right now include the following:

DHL logistic centre uses integrated AR smart glasses in the packing process to reduce lookup time, optimize navigational routes and guide the worker to the correct location.



Credit: Roman Zaiets / Shutterstock

FIGURE 6.27 Immersive Experiences for Entertainment

BMW uses AR headsets for workstation planning to enhance productivity, assess processes and foster training of engine units.

Adidas uses AR for imparting virtual employee training. 360° content is used to prepare employees for real-life situations (e.g., a Black Friday sale simulation).

Autodesk Revit Live and Iris VR provide immersive architectural and building visualizations to help clients visualize buildings even before starting with the construction phase.

6.12.4 Virtual Reality Systems

Virtual reality is a complete artificial world that does not exist at all. Users 'immerse themselves' into this non-existent environment not only as an observer but also as a participant. VR devices are used to move around and manipulate objects virtually. Some common characteristics of VR systems are as follows:

Real-time simulation. The VR system provides picture sound, and other sensations (if any) in response to the actions performed, instantly, with-

out noticeable delays.

Realistic imitation of the user's environment. To give an illusion of real-world to fully immerse the user in the world of virtual reality, the system should display virtual objects with realistic effects.

Multi-user mode. VR systems facilitates collaborative work by multiple users simultaneously.

Interactivity. The virtual world must work based on user's interaction with virtual objects. Here, the user is not just an extremely active observer but can also move, manipulate objects, look in different directions and perform other actions.

VR can be experienced through mobile devices. In 2014, Google invented that allows an immersive experience through a smartphone screen.

Some companies that have successfully used VR are as follows:

1. The Teleporter 4D launched by the Marriott hotel chain in 2014 provides virtual reality booths that take the user on an introductory tour of one of the cities. The booth also simulated the climatic conditions of the selected city.
2. Lexus facilitates users to take a virtual test drive of one of its latest models. The user just needs a smartphone and Google Cardboard.
3. Samsung introduced the 'Be Fearless' programme in 2016 to help users overcome their fears. Participants took a course using Samsung Gear VR to check if they are able to deal with stressful situations in virtual reality so that if they land up in a similar situation in real life, they can handle them in a better way without any phobia.
4. McDonald's introduced a new packaging for Happy Meal, which can be transformed into virtual reality glasses.

Advantages of VR

1. VR provides diverse types of instant data.

2. It presents images from multiple viewpoints.
3. VR demonstrates non-visible data to the user (like in geochemistry).
4. Facilitates users to ‘visit’ the places normally inaccessible to individuals.
5. VR provides an experience which can be recreated anytime.

Limitations of Virtual Reality

1. It only creates an imaginary world that has no connection with the real world.
2. VR cannot deal with real-world objects.
3. Though it can be used for educational purposes, it is seldom used as it devalues the importance of human connections and synergy in teaching-learning process.
4. VR systems are rigid and lacks flexibility

6.12.5 Types of Augmented Reality

Augmented reality is divided into four different categories as follows:

Marker based augmented reality uses interfaces like camera or some type of visual marker (like QR code) to produce result when the marker is sensed by a reader.

Marker-less augmented reality uses a GPS, digital compass, velocity/accelerometer which is embedded in devices to provide data based on user’s location.

Projection-based augmented reality projects artificial light on real world objects that senses human touch to exhibit a miracle.

Superimposition-based augmented reality partially or fully replaces the original view of an object with a newly augmented view of that same object.

Advantages of Augmented Reality

1. Increase sales: Apps such as lenskart are AR-based apps which allow users to check if the product looks good on him/her or not. This in-

creases customer interaction, satisfaction and ultimately boosts sales.
2. Enriches content: Unlike VR, AR facilitates interaction with the real world, which enriches experience and content.
3. For these reasons, AR is extensively being used for training and skill development applications in various industries (like military, pilot training, nuclear plant training, etc.).

Limitations of Augmented Reality

1. Sometimes, implementation of AR makes the application too ambiguous to use. When there are more layers, it becomes difficult to match with the exact environment and, therefore loses its user-friendly charm.
2. Using too much AR especially where it was not required, may obscure or narrow a user's interests or tastes.
3. Privacy control is a matter of concern with AR.

Therefore, some applications prefer to use MR rather than pure AR or VR.

6.12.6 Mixed Reality

Mixed reality is the combination of the best features of VR and AR. It ensures that limitations of both AR and VR are addressed to overcome them. MR integrates VR content with real time linear video and allows user interaction with virtual or real objects (refer **Fig. 6.28**).



Credit: Gorodenkoff / Shutterstock

FIGURE 6.28 Interacting with Objects using MR Techniques

Image Source: Microsoft

Thus, MR a kind of hybrid system that includes both physical and virtual elements. It is a right blend between a fully physical environment with no virtual elements, and a completely virtual environment. To use MR, users need to wear a headset that helps them see through a translucent viewport or glass on which visuals are projected.

AR visualization can create 3D catalogues and interactive demos of a product to make it visually appealing to customers.

MR tends to develop virtual characters for the real world to create a deeper sense of immersion for the user than AR alone does. Since MR facilitates interaction between virtual world and real-world objects in real-time to give a parallax effect, it is also referred to as augmented Reality 2.0 or hybrid reality. People can live, work, trek, shop and survive more realistically in an MR world than in VR.

Advantages of Mixed Reality

1. MR presents a highly interactive aspect.
2. MR renders realistic projection of virtual objects on the surroundings.

3. It allows users to interact with the immersive content using their body (like, finger gestures). It is not necessary to interact only with remote controllers or phone screens.

Disadvantages of Mixed Reality

1. The cost of MR devices is high.
2. There are issues related to large file sizes and low-resolution content.
3. Personal life may get affected due to addiction
4. Business users need a sound technical team to address issues in a timely manner.

Some Exclusive Applications of Mixed Reality

1. MR has the potential to take interactive product content management (IPCM) to the next higher level.
2. MR facilitates collaborative work, especially in a global team having members working from remote locations.
3. MR supports the manufacturing sector by helping it to create mock-ups of processes.
4. In the health sector, MR has made surgeries simpler and more flexible than before. Doctors use smart glasses to concentrate on details.
5. In education, MR can be used to switch from e-learning to simulation-based learning, thereby transforming modern education system.

6.12.7 Challenges of Immersive Experiences

Though an immersive experience seems to provide an ocean of exciting opportunities, there are certain technological shortcomings, primary health and safety factors associated with it that prevent its use at large. Some other challenges that act as a roadblock in the adoption of immersive technologies include the following:

1. Immersive devices pose a big threat to security. Since applications are monitored automatically, third-party users can hack the server of a particular company to steal crucial information (about data, finances, etc.) about a company.

2. The technology can be misused to perform crimes like cyber bullying, harassment, cyber stalking and leaking data of public figures.
3. An immersive experience impacts cognitive concentration that comes naturally to humans. It can fully replace a user's sensory connection to the real world.
4. Immersive experiences work best with a 5G Internet connection which is still not available everywhere.
5. The technology comes with a high cost.
6. Not every user has the digital knowledge to operate immersive devices, making immersive experience viable only for a limited set of users.



PROGRAMMER'S ZONE

Python program to demonstrate erosion and dilation of images.

```
''' Dilation adds pixels to the boundaries of objects in an image,
while erosion removes pixels on object boundaries.
Erosion is the counter-process of dilation. If dilation enlarges an
image then erosion shrinks the image'''
import cv2
import numpy as np
# Reading the input image
img = cv2.imread('result.jpg', 0)
# Taking a matrix of size 5 as the kernel
kernel = np.ones((5,5), np.uint8)
#The first parameter is the original image, # kernel is the matrix
with which image is convolved and # third parameter is the number of
iterations, which will determine how much you want to #erode/dilate
a given image.
img_erosion = cv2.erode(img, kernel, iterations=1)
img_dilation = cv2.dilate(img, kernel, iterations=1)
cv2.imshow('Input', img)
cv2.imshow('Erosion', img_erosion)
cv2.imshow('Dilation', img_dilation)
cv2.waitKey(0)
```



Credit: Screenshot used with permission from Microsoft

HANDLING MOUSE EVENTS

```
import cv2

# read image
img = cv2.imread('result.jpg')
# show image
cv2.imshow('image', img)

# define the events for the mouse_click.
def mouse_click(event, x, y,flags, param):

  # to check if left mouse button was clicked
  if event == cv2.EVENT_LBUTTONDOWN:

    # font for left click event
    font = cv2.FONT_HERSHEY_TRIPLEX
    LB = 'Left Button'

    # display that left button was clicked.
     # 1 is fontScale that is multiplied by the font-specific base
size.
    # 2 is the thickness of the line in px.
    cv2.putText(img, LB, (x, y), font, 1, (255, 255, 0), 2)

    cv2.imshow('image', img)
    # to check if right mouse button was clicked
    if event == cv2.EVENT_RBUTTONDOWN:

      # font for right click event
      font = cv2.FONT_HERSHEY_SCRIPT_SIMPLEX
      RB = 'Right Button'

      # display that right button was clicked.
      cv2.putText(img, RB, (x, y), font, 1, (0, 255, 255), 2)
      cv2.imshow('image', img)
  cv2.setMouseCallback('image', mouse_click)

  cv2.waitKey(0)
  # close all the opened windows.
  cv2.destroyAllWindows()
```



Generating image pyramid

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread("result.jpg")
layer = img.copy()
for i in range(4):
  '''The layout is organized in rows and columns, which are represented
by the first and second argument. The third argument represents the
index of the current plot.'''
  plt.subplot(2, 2, i + 1)
  # using pyrDown() function
  '''The pyrUp() function increases the size to double of its original
size and pyrDown() function decreases the size to half.'''

  layer = cv2.pyrDown(layer)
  plt.imshow(layer)
  cv2.imshow(str(i), layer)
  cv2.waitKey(0)
  cv2.destroyAllWindows()
```



DRAWING RECTANGLE

```
import cv2
import numpy as np
draw = False # true if the mouse is pressed. Press m to shift into
curve mode.
#mode = True # if True, draw rectangle.
a,b = -1,-1
# mouse callback function
def draw_rect(event,x,y,flags,param):
  global a,b,draw,mode
  if(event == cv2.EVENT_LBUTTONDOWN):
    draw = True
    a,b = x,y
  elif (event == cv2.EVENT_MOUSEMOVE):
    if draw == True:
 #      if mode == True:
      cv2.rectangle(img,(a,b),(x,y),(0,255,0),-1)

  elif(event == cv2.EVENT_LBUTTONUP):
    draw = False

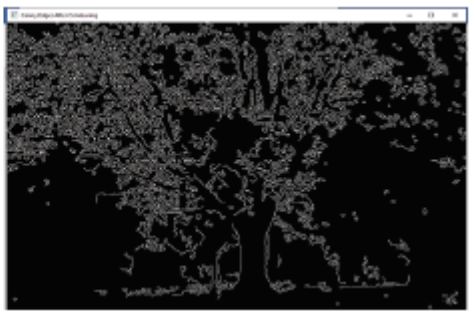
# We bind the keyboard key m to toggle between rectangle # and circle.
img = np.zeros((512,512,3), np.uint8)
cv2.namedWindow('image')
cv2.setMouseCallback('image',draw_rect)
while(1):
  cv2.imshow('image',img)
  k = cv2.waitKey(1)
  if(k == 27):
    break
cv2.destroyAllWindows()
```



DRAWING CONTOURS

```
import cv2
import numpy as np
# Let's load a simple image with 3 black squares
image = cv2.imread('result.jpg')
cv2.waitKey(0)
# Grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Find Canny edges
'''Image: Input image to which Canny filter will be applied
T_lower: Lower threshold value in Hysteresis Thresholding
T_upper: Upper threshold value in Hysteresis Thresholding'''
edged = cv2.Canny(gray, 30, 200)
cv2.waitKey(0)
# Finding Contours
# Use a copy of the image e.g. edged.copy()
# since findContours alters the image
'''Contours are defined as the line joining all the points along the
boundary of an image that are having the same intensity.
If we pass cv2.CHAIN_APPROX_NONE, all the boundary points are stored.
If we have to find the contour of a straight line then use cv2.
CHAIN_APPROX_SIMPLE does
cv2.RETR_EXTERNAL considers only the extreme outer contours'''
contours, hierarchy = cv2.findContours(edged, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_NONE)
cv2.imshow('Canny Edges After Contouring', edged)
cv2.waitKey(0)
print("Number of Contours found = " + str(len(contours)))
# Draw all contours
# -1 signifies drawing all contours and 3 is the thickness. Usually,
it is set to 0
cv2.drawContours(image, contours, -1, (0, 255, 0), 3)
cv2.imshow('Contours', image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```







KEY TERMS

Computer vision: A field of artificial intelligence that helps computers gain a high level of understanding from digital images or videos.

Convolutional Neural Network (CNN): A branch of deep learning that uses multi-layered neural networks to analyse visual imagery. For this, CNN uses a special technique called convolution.

Convolution: A mathematical operation on two functions that produces a third function that expresses how the shape of one is modified by the other.

Object Classification: The process of identifying the broad category of object in the photograph.

Object Identification: The process of classifying the type of a given object is in the photograph.

Object Verification: The process of confirming the presence of an object in the photograph.

Object Localization: The process of locating objects in the photograph. The goal of **localization** is to find the location of a single object in an image.

Object Detection: The process of repeatedly locating and classifying all objects of interest in the image.

Object Segmentation: The process of enumerating the pixels that belong to an object in the image.

Object tracking: The process of detecting an object and then discern its movements.

Video motion analysis: The process of estimating the velocity of objects in a video, or the camera itself.

Scene reconstruction: The process of creating a 3D model of a scene inputted through images or video.

Image restoration: The process of using machine learning based filters to remove noise (to remove parts of image that are blurred).

Visual Search Engines: A search engine that retrieves images meeting the specified criteria.

Noisy image: An image with incomplete/incorrect data makes it hard for computer vision to recognize objects.

Adversarial Attacks: Attacks in which an attacker intentionally attacks a machine learning model to make the machine fail. These flawed models are difficult to identify and may affect the system adversely.

Immersive Experience: A technology that extends reality or creates a new reality by leveraging the 360-degree space.

360-degree content: An interactive photo or a video recording shot in every direction at the same time. This allows users to change the viewing direction at any moment.

Virtual reality (VR): A computer simulation that immerses the user inside a virtual world and allows him to interact with the environment.

Augmented reality (AR): Technology that enhances the user's perception of the real world by adding a computer-simulated layer of information on top of it.

Digital Twins: Near-exact virtual models of real-life objects, processes or systems that is usually used in manufacturing or engineering to simulate physical things either for optimizing or for studying how they behave before actually building them.



RECAPITULATION POINTS

- A CNN complements the machine learning or deep learning model by breaking images down into pixels that are given tags or labels.
- The black-and-white image stores a '0' for a black pixel and a '1' for white pixel. This is known as a 'bitmap' image as it only requires one bit of information, 0 or 1, to describe the color

- Computer vision algorithms can be used for predictive maintenance. Images obtained through cameras (probably fitted in the eyes of robots) can be analysed to detect potential problems they occur. CV can even be helpful to reduce defects.
- CNNs are a special type of neural network that roughly imitate human vision.
- Edge detection is an image processing technique used to identify object's boundaries within an image.



REVIEW QUESTIONS

1. Discuss the applications and challenges of computer vision.
2. CNN is a composition of three main types of layers. Elaborate this statement.
3. What is the role of activation function in a neural network and in a CNN?
4. Give the different elements of immersion.
5. Write a short note on mixed reality.
6. What do you understand by the term 'computer vision'?
7. Explain the tasks involved in computer vision.
8. Computers can detect objects better than humans. Justify this statement.
9. How does computer vision work with deep learning?
10. How can we use computer vision in manufacturing units?
11. Give any four areas where CV is extensively used.
12. List any three challenges faced by computer vision systems.
13. What do you understand by CNN? Explain the different layers that makes up a CNN model.
14. Differentiate between CNN and ANN.
15. Differentiate between bitmap and a greyscale image.
16. What is image resolution? Discuss two ways in which resolution can be specified.
17. Give any two benefits of using an activation map in a CNN.
18. Why do we need multiple convolution layers?
19. Differentiate between
    1. Image size and image resolution

2. CNN and ANN
3. Augmented and virtual reality



FILL UP

1. _____ systems like Google Lens identify images by identifying image properties like composition, colour, texture, etc.
2. The word _____ means ‘picture element’ indicating one single dot.
3. _____ allows users to change the viewing direction at any moment.
4. Near-exact virtual models of real-life objects, processes or systems are known as _____.
5. Image _____ is an image processing technique that removes noise from an image.
6. Auto companies usually use _____ to perform a dummy crash test in a metaverse world.
7. A _____ neural network is used in computer vision algorithms.
8. _____ is the process of classifying the type of a given object is in the photograph.
9. Object _____ is performed to detect an object and then discern its movements.
10. _____ systems recognize faces in images and videos.
11. Pokemon Go and Snapchat use _____ reality applications.
12. A _____ search engine retrieves images that meets the specified criteria.
13. _____ attacks are intentionally done on a machine learning model to make the machine fail.
14. _____ neural networks are a special type of neural network that roughly imitate human vision.
15. _____ is one of the most popular libraries providing functions for computer vision.
16. Tiny dots on a monitor are known as _____.
17. _____ refers to the detail that an image holds.



STATE TRUE OR FALSE

1. Image restoration uses machine learning-based filters to remove noise.
2. A 300 PPI image will also be a 300 DPI image.
3. CNN has only one convolutional layer.
4. Average pooling performs better than max pooling.
5. Computer vision makes extensive use of deep learning techniques.
6. To remove noise from an image, we use scene reconstruction technique.
7. Computer vision algorithms can detect diseases by analysing X-rays, MRI and mammography.
8. Google Lens is used to obtain detailed information from images.
9. A fully connected layer is arranged next to series of convolution and pooling layer.
10. The pooling layer is followed by a convolution layer.
11. Average pooling performs better than average pooling.
12. An image is rotated in multiples of 90 degrees.
13. Fully connected layers are present in a CNN.
14. Pooling layer is present in a CNN but not in ANN.
15. Pixel means picture element.
16. In a bitmap, every pixel stores the value of RGB components.
17. A bitmap cannot represent shades of grey.
18. Fully connected layers are present in a CNN.
19. Pooling Layer is present in a CNN but not in ANN.
20. Pixel means picture element.
21. In a bitmap, every pixel stores the value of RGB components.
22. A bitmap cannot represent shades of grey.
23. Higher the resolution of image, lesser is its clarity.
24. A 1280 X 1024 pixels camera has a higher resolution than a 3 mega pixel camera.



MULTIPLE CHOICE QUESTIONS

1. Object _____ finds and classifies a *variable number of objects* in the given image.
    1. Verification
    2. Detection
    3. Localization
    4. Classification
2. A greyscale image can represent a total of _____ different colours.
    1. 1
    2. 2
    3. 256
    4. 16 million
3. Image _____ is often measured in Dots Per Inch (DPI) and Pixels Per Inch (PPI).
    1. Hue
    2. Resolution
    3. Intensity
    4. Size
4. Which layer of convolution neural network reduces the size of inputs to speedup computations?
    1. Convolution
    2. Pooling
    3. Fully-connected
    4. All of these
5. _____ enhances the user's perception of the real world by adding a computer-simulated layer of information on top of it.
    1. Augmented reality
    2. Virtual Reality
    3. 360 -degree content
    4. Digital Twin
6. _____ reality integrates virtual objects in the real world.
    1. Augmented
    2. Virtual
    3. Mixed
    4. All of these

7. _____ is an image smoothening technique.
    1. Averaging
    2. Gaussian Blur
    3. Bilateral Filter
    4. All of these
8. The size of an image is represented as 600 x 300 x 3. How many channels does it have?
    1. 600
    2. 300
    3. 3
    4. 0
9. The size of an image is represented as 600 x 300 x 3. How many pixels does it have in a single row?
    1. 600
    2. 300
    3. 3
    4. 0
10. The size of an image is represented as 600 x 300 x 3. How many pixels does it have in a single column?
    1. 600
    2. 300
    3. 3
    4. 0
11. The size of an image is represented as 600 x 300 x 3. What would be the type of the image?
    1. Bitmap
    2. Grayscale
    3. Color
    4. None of the above
12. Object _____ technique is used by news channel to blur the face of a victim or any person whose identity is not to be revealed?
    1. Identification
    2. Segmentation
    3. Verification
    4. Localization
13. The process of repeatedly locating and classifying all objects of interest in the image is known as object _____.

    1. Verification
    2. Detection
    3. Localization
    4. Segmentation

14. The process of creating a 3D model of a scene inputted through images or video is known as _____.
    1. Scene reconstruction
    2. Object tracking
    3. Video motion analysis
    4. Image restoration
15. _____ is used to estimate the velocity of objects in a video, or the camera itself.
    1. Scene reconstruction
    2. Object tracking
    3. Video motion analysis
    4. Image restoration
16. Computer vision cannot be used for _____ systems.
    1. Home security
    2. Augmented reality
    3. Speech recognition
    4. Gaming
17. Video cameras placed in/on _____ produce a noisy image.
    1. Dark room
    2. A road with heavy raining
    3. Bright sun
    4. All of these
18. Overlapping of image occurs when an image has _____.
    1. Missing data
    2. Different lighting
    3. One object hidden below another
    4. Been taken from different angles
19. CNN is a branch of _____ learning.
    1. Deep
    2. Supervised
    3. Unsupervised
    4. Reinforcement

20. _____ layer reduces the size of inputs by removing unnecessary information to speedup computations.
    1. Convolution
    2. Pooling
    3. Fully connected
    4. All of these
21. _____ layer performs computational tasks.
    1. Convolution
    2. Pooling
    3. Fully connected
    4. All of these
22. Which function is used to change the size of an image?
    1. imread()
    2. resize()
    3. rotate()
    4. size()
23. Which function is used to print the size of an image?
    1. shape()
    2. resize()
    3. rotate()
    4. size()
24. Which function is used to print the number of rows, columns and colours?
    1. shape()
    2. resize()
    3. rotate()
    4. size()
25. Which technique removes noise from an image?
    1. Blurring
    2. Smoothing
    3. Detecting edges
    4. Resizing
26. Which technique used to hide the details or any confidential information in an image?
    1. Blurring
    2. Smoothing
    3. Detecting edges

    4. Resizing
27. An ANN does not have _____ layer.
    1. Input
    2. Output
    3. Convolution
    4. Hidden
28. Which of the following statements is incorrect about a CNN?
    1. CNN is more efficient than ANN and has a higher success rate.
    2. CNN requires for data for training than an ANN.
    3. CNN uses a greater number of parameters for learning than an ANN.
    4. Computational cost of CNN is more than that of ANN.
29. How many different colours can a greyscale image store?
    1. 2
    2. 256
    3. 20
    4. 16,535
30. A _____ image stores either of two values in a pixel.
    1. Bitmap
    2. Greyscale
    3. Coloured
    4. All of these
31. An ANN does not have _____ layer.
    1. Input
    2. Output
    3. Convolution
    4. Hidden
32. Image _____ is often measured in Dots Per Inch (DPI) and Pixels Per Inch (PPI).
    1. Width
    2. Height
    3. Resolution
    4. Brightness
33. An image with resolution 2048 x 1536 has _____ number of pixels in a row and _____ pixels in a column.
    1. 2048, 1536
    2. 1536,2048

3. 3 mega pixels
4. All of these

If the values of two neighbouring squares on the grid differ by more than 50, draw a thick line between them. (Taken from NCERT. Can add 10 to all the numbers).

21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21
21	21	21	21	21	21	21	103	103	21	21	21	21	21	21	21
21	21	21	21	21	21	21	99	103	21	21	21	21	21	21	21
21	21	21	21	21	21	21	91	103	21	21	21	21	21	21	21
21	21	21	21	21	21	21	103	99	21	21	21	21	21	21	21
21	21	21	21	21	21	91	99	103	99	21	21	21	21	21	21
21	72	93	101	103	105	99	103	91	99	104	102	101	91	103	21
21	21	99	99	91	103	99	99	103	99	99	91	99	88	21	21
21	21	21	99	99	91	91	99	103	99	91	99	103	21	21	21
21	21	21	21	99	99	99	99	99	99	99	103	21	21	21	21
21	21	21	21	21	99	103	103	91	103	103	21	21	21	21	21
21	21	21	21	21	103	91	99	99	103	91	21	21	21	21	21
21	21	21	21	103	91	99	99	103	99	103	99	21	21	21	21
21	21	21	21	91	103	99	21	21	99	91	99	21	21	21	21
21	21	21	103	103	21	21	21	21	21	21	103	99	21	21	21
21	21	21	103	3	21	21	21	21	21	21	7	103	21	21	21

If the values of two neighbouring squares on the grid differ by more than 40, draw a thick line between them.

IDENTIFY THE PROCESS



Credit: Redchanka / Shutterstock

_____



Credit: Zapp2Photo / Shutterstock

_____



Credit: Tsomka / Shutterstock

_____

ANSWERS

Fill Up

1. Content-based image retrieval
2. ‘pixel’
3. 360-degree content
4. Digital twins
5. smoothing
6. Digital twins
7. Convolutional

8. Object identification
9. tracking
10. Facial recognition
11. Augmented
12. visual
13. Adversarial
14. Convolutional
15. OpenCV
16. pixels
17. Resolution

State True or False

1. True
2. True
3. False
4. False
5. True
6. False
7. True
8. True
9. True
10. False
11. False
12. True
13. False
14. True
15. True
16. False
17. True
18. False
19. True
20. True
21. False
22. True
23. False
24. False

Multiple Choice Questions

1. b
2. c
3. b
4. b
5. a
6. c
7. d
8. c
9. a
10. b
11. c
12. b
13. c
14. a
15. c
16. c
17. d
18. c
19. a
20. b
21. a
22. b
23. d
24. a
25. b
26. a
27. c
28. c
29. b
30. a
31. c
32. c
33. a